Reproducible environment and records

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Reproducible environment and records before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Power and resource tradeoffs

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predefining an analysis plan

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset and benchmark limitations

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Reproducible environment and records trace and record the first step you can explain without help.

Use W2: Trace the worked case for Reproducible environment and records. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Reproducible environment and records explanation against the supplied facts before discussing it.

Use the worked case for Reproducible environment and records. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> Partner changes pre**process**ing and data version while keeping old run ID. What must be preserved?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Reproducible environment and records facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Reproducible environment and records facts and keep your reasoning.

Folder contains seed42 and final-score.txt, but no dataset, code or dependency record. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Reproducible environment and records facts and keep your reasoning.

A report omits this qualification: Reproducibility supports checking the computation, not automatic validity of the research claim.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Reproducible environment and records facts and keep your reasoning.

Partner changes pre**process**ing and data version while keeping old run ID. What must be preserved?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

Design a revealing follow-up. Apply the supplied Reproducible environment and records facts

> and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Reproducible environment and records to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled reproducible environment and records **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reproducible environment and records.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

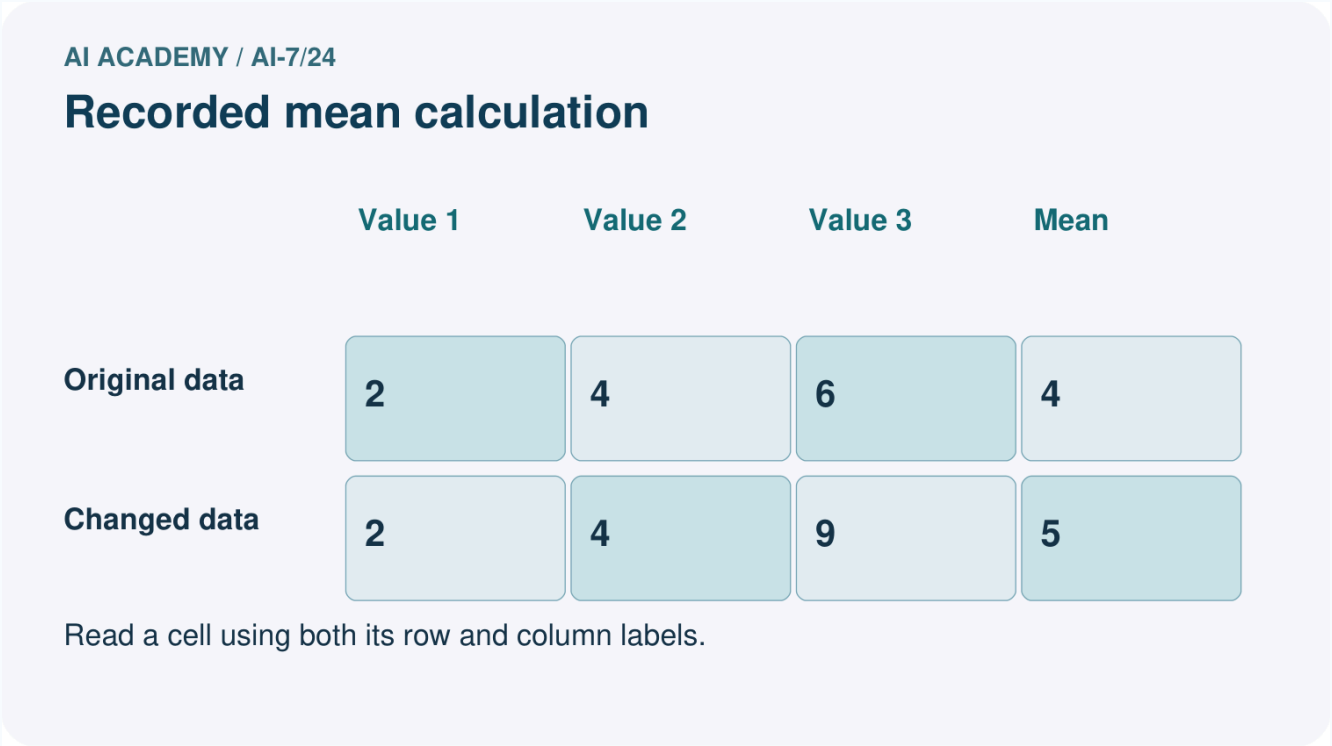
WORKBOOK · 12 OF 15

Pictorial case practice: Reproducible environment and records

> A reproducible calculation record. A teammate must rerun a three-value mean without guessing the input or software version. Explain the labelled step before checking the changed condition.

A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version.

Data file contains 2,4,6. Recorded script sums values and divides by count; environment and file hash are saved.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/24

Reproducible environment and records

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify exact input

Use the three recorded values, not a similarly named edited file.



2. Trace calculation

Sum is 12; count is 3; mean is 4.



3. Save provenance

Retain script version, environment description and data hash with the output.

CHANGE ONE THING

The data file changes to 2,4,9.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Retain script version, environment description and data hash with the **output**.

1. Trace calculation

2. Identify exact **input**

3. Save provenance

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Save provenance Retain script version, environment description and data hash with the **output**.

Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The data file changes to 2,4,9. Which conclusion is supported by the supplied facts?

1. The mean becomes 15/3=5, and the saved **input** hash must change.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The mean becomes 15/3=5, and the saved **input** hash must change. The calculation is the same but its **input** records differ.

Repair a missing building block

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

Explain each Reproducible environment and records picture from its labels, then record a relevant limit.

01. Reproducible environment and records: the case

AI ACADEMY / AI-7/24 · 01 · LOOK AT THE FACTS

Reproducible environment and records: the case

The situation

A reproducible calculation record. A teammate must rerun a three-value mean without guessing the input or software version.

Supplied evidence

Data file contains 2,4,6. Recorded script sums values and divides by count; environment and file hash are saved.

Explain reproducible environment and records using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

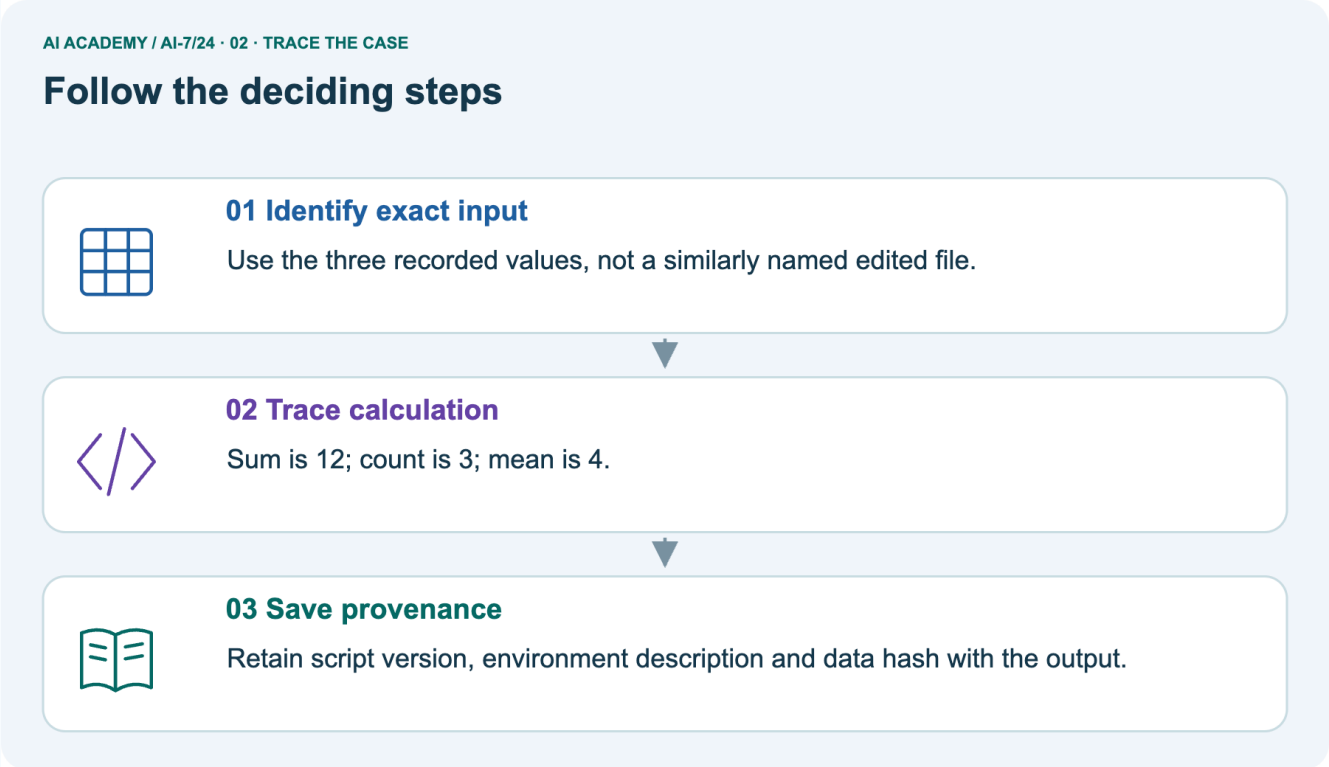
A reproducible calculation record. A teammate must rerun a three-value mean without guessing the **input** or software version. Supplied facts: Data file contains 2,4,6.

Recorded script sums values and divides by count; environment and file hash are saved.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



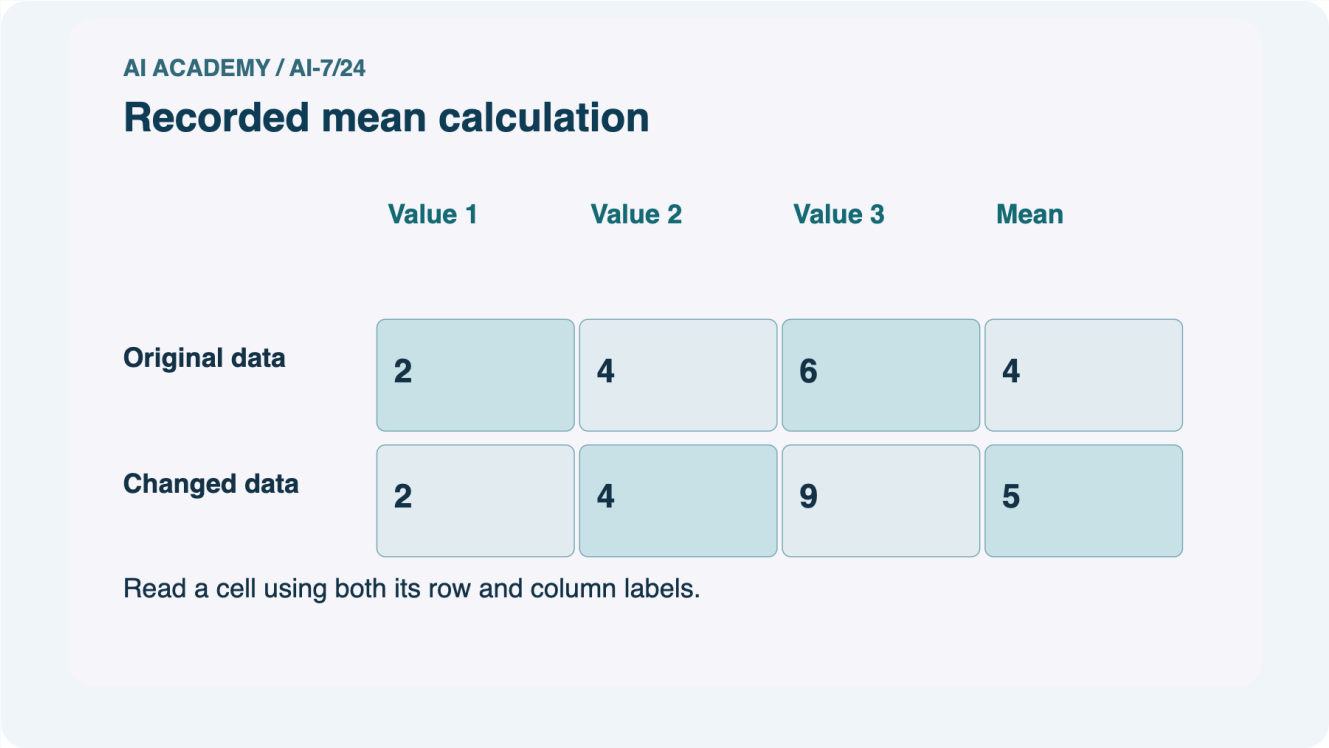
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify exact **input**: Use the three recorded values, not a similarly named edited file. → Trace calculation: Sum is 12; count is 3; mean is 4. → Save provenance: Retain script version, environment description and data hash with the **output**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Recorded mean calculation



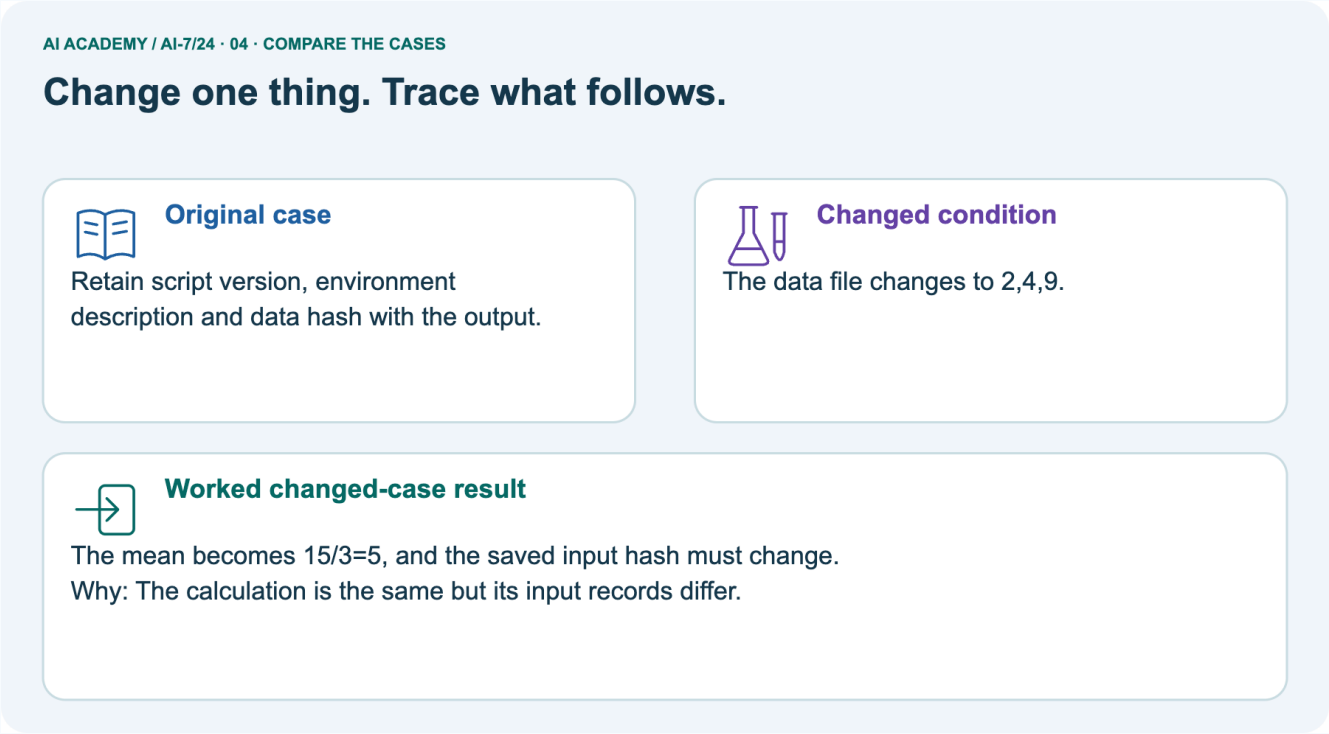
Follow the labelled picture with the matching explanation. Open the vector diagram

Recorded mean calculation. Original data: Value 1 = 2, Value 2 = 4, Value 3 = 6, Mean = 4. Changed data: Value 1 = 2, Value 2 = 4, Value 3 = 9, Mean = 5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

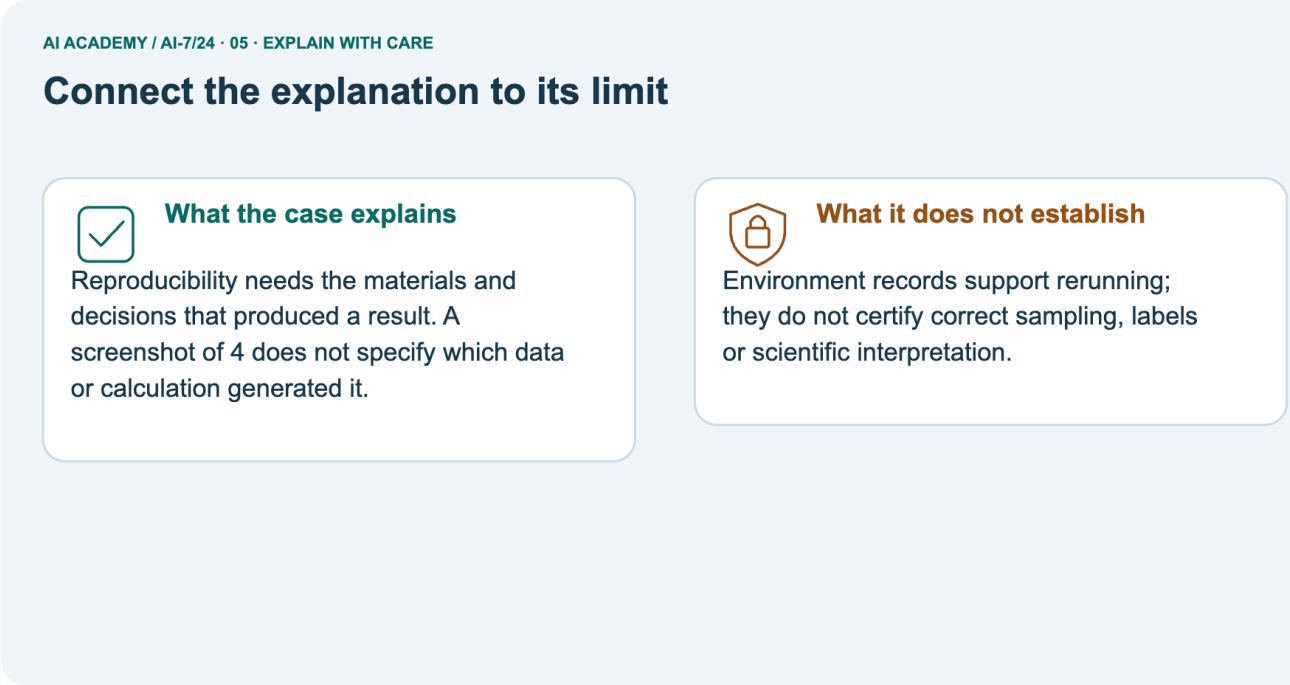
Original case: Retain script version, environment description and data hash with the **output**. Changed condition: The data file changes to 2,4,9. Expected result: The mean becomes 15/3=5, and the saved **input** hash must change.

Reason: The calculation is the same but its **input** records differ.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

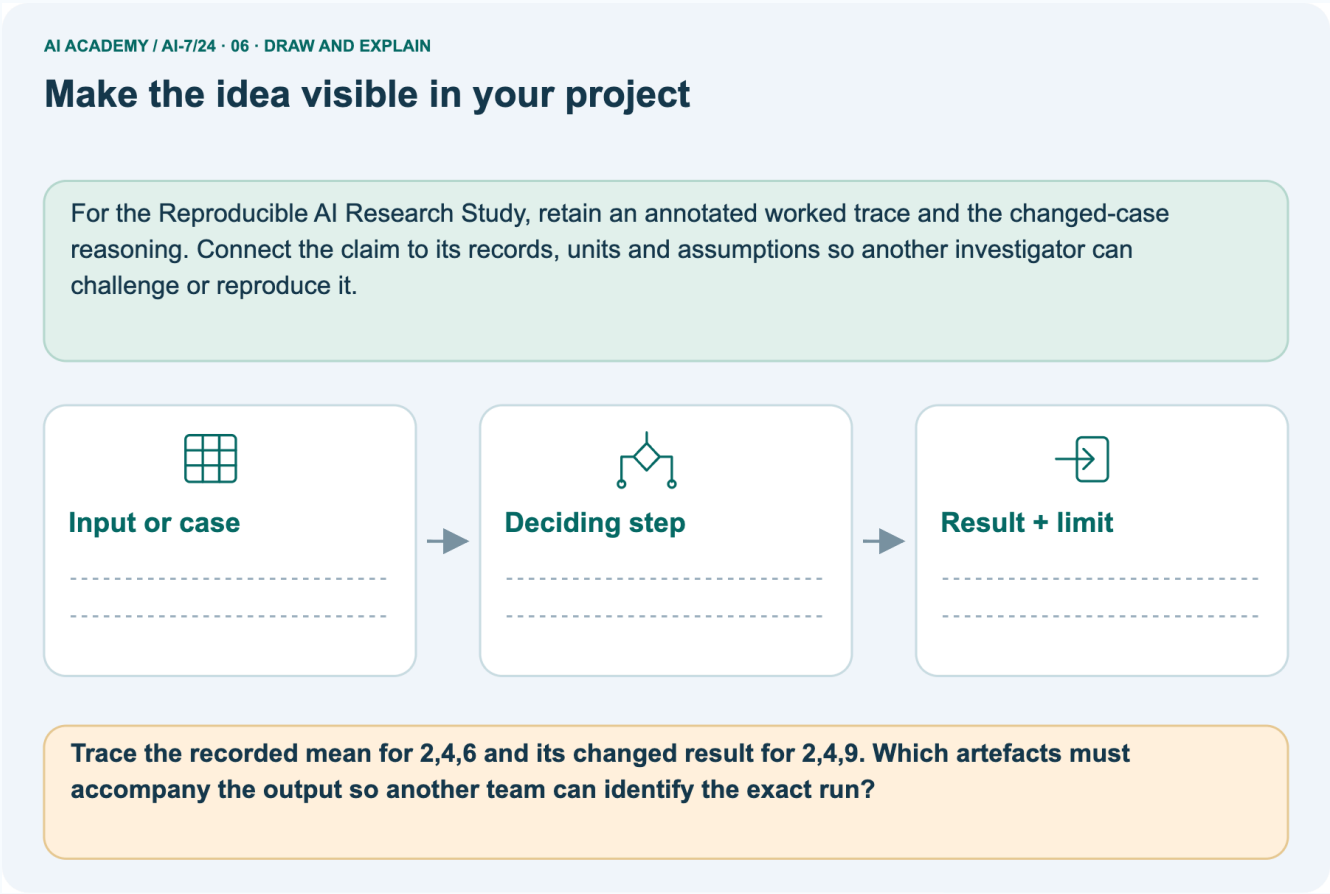
Reproducibility needs the materials and decisions that produced a result. A screenshot of 4 does not specify which data or calculation generated it.

Limit: Environment records support rerunning; they do not certify correct sampling, labels or scientific interpretation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the **output** so another team can identify the exact run?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.